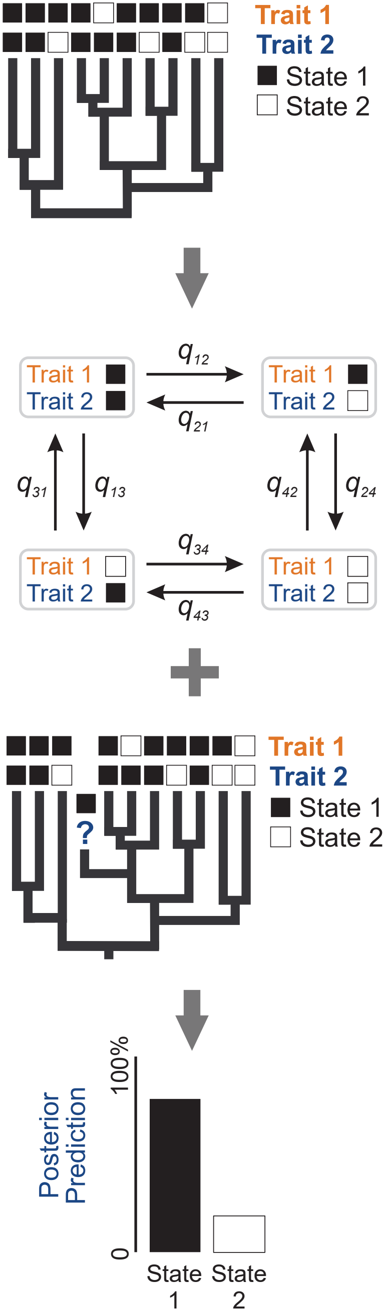

Step 1: Collect discrete data for taxa. This workflow assumed that two co-evolving characters are under study. The same procedure holds for a single discrete trait. For taxa that have missing data score their trait states as "-". The phylogeny can have different types of branch lengths (time vs. molecular) and a posterior distribution of trees can be used to account for phylogenetic uncertainty.

Step 2: Produce a posterior distribution of character rate change models between using a reversible-jump Markov Chain Monte Carlo (MCMC). This will automatically reduce the number of parameters and estimate if the traits co-evolved together. Taxa with "-" for trait values will not be used to estimate these models.

Step 3: Re-run the analysis, but load the models from step 2 and change "-" to "?".

Step 4: The trait states are predicted for taxa with missing data using the models of evolution and the location of the taxa on the tree.